
Async LSP-Supervised Decoding for Code Generation

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Abstract

Verifier-guided code-generation systems — Monitor-Guided Decoding [1], ROCODE [12], SemGuard [22], and Type-Constrained Code Generation [16] — all share one design choice: the verifier runs *synchronously* on the decoding critical path. The decoder stalls while the language server, compiler, or learned classifier returns its verdict. This is tolerable when the verifier is fast (Python’s `compile()`, `g++ -fsyntax-only`, or a 1.3B classifier on one line of code), but it forecloses the use of slower, sounder oracles — `cargo check` on Rust (~ 100 ms to several seconds), `javac` on Java (~ 180 ms), or whole-crate type and lifetime analyses. We present *SoundCode*, a producer-consumer decoding loop in which a Language Server Protocol (LSP) [1, 2] verifier runs *asynchronously* alongside the LLM, spawning a check at each depth-zero structural boundary (`;` or `}`) and triggering rollback to the last verified checkpoint only when a returned diagnostic is blocking. The same rollback algorithm underlies the synchronous ROCODE baseline [12]; we isolate the sync-versus-async axis at fixed verifier, fixed model, and fixed benchmark. We evaluate Qwen2.5-Coder-7B-Instruct [10] on MultiPL-E HumanEval-rs and HumanEval-cpp [4]. Holding the model, the verifier, and the rollback algorithm fixed, SoundCode is $1.59\times$ faster on Rust (116/156 paired wins) and $2.95\times$ faster on C++ (159/161 paired wins) than the synchronous ROCODE-style equivalent, while *raising* C++ pass@1 from 71.4% to 76.4%. Beyond the measurement, we offer a design framework — when to async, when to roll back, and where to roll back — that applies to any verifier whose cost is not negligible relative to a single token’s decoding latency. A live web demo and reference implementation are open-source.

1 Introduction

Large language models predict next tokens; they do not enforce programming-language rules. Even after billions of training tokens of high-quality code, instruction-tuned code LMs hallucinate ill-typed expressions, undefined identifiers, ill-formed borrows, and nonsensical control flow. The hallucinations are not bugs in particular models — they are a structural consequence of how autoregressive sampling commits to one token at a time without look-back: any path the model places non-trivial probability mass on becomes reachable, and a single bad token can cascade [12, 22]. Recent work on the *intrinsic* sources of hallucination in language modelling confirms this: a frozen LM that has been trained on data with even a small fraction of subtly incorrect programs will reliably generate some subtle errors at inference, because next-token cross-entropy gives no mechanism to refuse paths that violate global constraints [17].

Programming-language verifiers are exactly the missing mechanism. A compiler, a type checker, a borrow checker, or a Language Server Protocol (LSP) server, taken on the decidable fragment of

*Code and live demo: github.com/leobwang/soundcode; soundcode-demo.psixyzt.com.

code it covers — syntax, type, name resolution, lifetimes — returns a ground-truth verdict: either the program is well-formed or it is not. This verdict is decoupled from how the program was produced. It is also exactly the kind of side information a generative LM has no internal way to consume. The natural design move is to put one in the decoding loop.

Several recent systems do exactly this. Monitor-Guided Decoding (MGD; 1) consults an LSP at hand-picked syntactic triggers — the `.` that begins a member-access expression in Java — and masks the LM’s next-token logits to those prefixing a type-consistent identifier. ROCODE [12] runs the compiler incrementally after each generated statement, rolls back the buffer on an error, and decodes the retry under an exponentially-decaying token-penalty mask. SemGuard [22] swaps the compiler for a learned 1.3B semantic classifier that scores each emitted line and triggers line-level rollback below a threshold. Type-Constrained Code Generation [16] formalizes per-token soundness in TypeScript by composing a prefix automaton with a type-reachability search.

All four share one design choice: the verifier runs *synchronously* on the decoding critical path. The LM stops; the verifier returns; the LM resumes. This is invisible when the verifier is fast — Python’s `compile()` parses a partial program in ~ 0.1 ms, `g++ -fsyntax-only` takes ~ 3 ms, and a 1.3B classifier evaluates one line of context in single-digit milliseconds on a modern GPU. ROCODE’s benchmarks are Python and C++ [12], SemGuard’s are Python and Java [22], MGD’s are Java with hand-picked dereference triggers [1]. In each case the per-call verifier cost is well below the LM’s own per-token latency, so the slowdown is tolerable. MGD reports a $\sim 83\%$ mean-decoding-time slowdown [1]; Type-Constrained reports a 39–52% median per-instance overhead [16]. The cost is real but amortized away by the gain in code quality.

The same design fails to scale to slower oracles. `cargo check` on a typical Rust function takes ~ 100 to several hundred milliseconds; `javac` on a single Java method takes ~ 180 ms once JVM startup is amortized; a whole-crate borrow check on an unfamiliar trait expansion can take seconds. A synchronous decoding loop that blocks on such an oracle spends most of its wall time idle. The two natural responses — (1) substitute a faster, weaker oracle, e.g. a learned classifier [22]; or (2) accept a $> 10\times$ slowdown and amortize against the larger code-quality gain — both surrender something the design originally promised. The first sacrifices soundness (a learned classifier is wrong on a defined fraction of inputs by construction); the second sacrifices the wall-clock budget the verifier-guided pipeline was meant to outperform.

This paper takes a different response: *run the verifier asynchronously*. We introduce **SoundCode**, a producer-consumer decoding loop in which:

- the LM streams tokens uninterrupted (the producer);
- at each depth-zero structural boundary — `;` or `}` at brace-depth zero in C-family languages, newline at indent zero in Python — a background task is spawned to run the verifier on the current buffer snapshot;
- a consumer task drains finished checks in FIFO order;
- on the first *blocking* diagnostic, pending checks are cancelled, the buffer is truncated to the most recent verifier-blessed checkpoint, and the LM is re-prompted from there.

Crucially, the same rollback algorithm underlies the synchronous ROCODE-style baseline that we compare against; *the only architectural difference is whether the verifier blocks the decoder*. This isolates the sync-vs-async axis at fixed verifier, fixed model, and fixed benchmark.

We evaluate SoundCode against ROCODE-style synchronous verification on HumanEval-rs ($n = 156$ after filtering) and HumanEval-cpp ($n = 161$) from MultiPL-E [4], using Qwen2.5-Coder-7B-Instruct [10] via vLLM. We report pass@1, mean wall-clock time per problem, mean rollback count, and compile rate, plus an ablation on rollback target (naive last-checkpoint vs. ROCODE’s max-entropy formula).

Headline result. Holding the model, the verifier, and the rollback algorithm fixed, SoundCode is $1.59\times$ faster on Rust (1.30 s sync $\rightarrow 0.82$ s async; $1.48\times$ at the per-problem median; async beats sync on 116 of 156 paired problems, 74.4%) and $2.95\times$ faster on C++ (3.65 s $\rightarrow 1.23$ s; $2.84\times$ at the median; async wins on 159 of 161 paired problems, 98.8%) than the synchronous ROCODE-style equivalent. The C++ result is the more interesting story: async *simultaneously* speeds the system up and *raises* pass@1 from 71.4% to 76.4% (+5.0 pp). On Rust, sync still leads pass@1 by 3.2 pp

(72.4% vs 69.2%), which we read not as a deficit of async but as a benchmark in which the cost-spectrum prediction (§3.3) places Rust near the marginal regime where sync’s mid-decode aborts cost less wall-time than its information value.

Contributions. We frame the contribution along four axes:

1. **The first asynchronous design for verifier-guided code generation.** Every prior boundary-rollback system [12, 22, 23] runs its verifier synchronously; every prior asynchronous-parallel-verify system [14, 5, 8] uses the LM itself as the verifier and gives up soundness. SoundCode occupies the open “semantic-verifier $\times$ structural-rollback $\times$ async” design point.
2. **A controlled head-to-head measurement.** At fixed model, fixed verifier, and fixed rollback algorithm — so the only axis that varies is whether the verifier blocks the decoder — async delivers a $1.59\times$ wall-time speedup on Rust (116/156 paired wins) and a $2.95\times$ speedup on C++ (159/161 paired wins), with C++ pass@1 simultaneously rising from 71.4% to 76.4%.
3. **A verifier-cost-spectrum framing that predicts when async helps.** We sort verifiers along a cost spectrum (Python’s `compile()` ~ 0.1 ms; `g++ -fsyntax-only` ~ 3 ms; `cargo check` ~ 35 ms; `javac` ~ 180 ms; whole-crate analyses up to seconds) and argue that the async design pays off whenever per-call verifier cost exceeds per-token decoding latency. The Rust-vs-C++ asymmetry in our results — a smaller win on the cheaper-verifier language — is the first empirical confirmation of this prediction.
4. **A reproducible open-source implementation and live demo.** The producer is vLLM’s AsyncLLMEngine [13] run with `enforce_eager=False`; the verifier is rust-analyzer via `multitlsy` [1, 2] or `cargo check / g++ -fsyntax-only` via subprocess. Engineering effort to port to a new language is one boundary-detector and one verifier-driver class. Source: `github.com/leobwang/soundcode`; live demo: `soundcode-demo.psixyzt.com` [25].

The Olausson et al. matched-budget framework [17] sets the bar any verifier-guided system must clear: a self-repair loop only beats matched-budget i.i.d. resampling when the feedback signal is *strictly stronger* than the generator. We argue that a real compiler is exactly that. A learned classifier is not. SemGuard [22] acknowledges this trade-off explicitly. Our async design lets us pay for the strictly-stronger oracle in wall-clock terms without paying for it in throughput.

2 Related Work

SoundCode sits at the intersection of three lines of work: (i) decoding-time methods that consult a programming-language verifier and react to its output — the closest neighbours, treated in detail below; (ii) per-token constrained-decoding methods that mask the LM’s logit vector against a grammar or type lattice but do not roll back; and (iii) speculative-decoding methods that run a verifier asynchronously alongside the LM, but use the LM itself as the verifier. SoundCode borrows the rollback algorithm from (i), the mask-when-cheap idea from (ii), and the parallel-verify control flow from (iii); its open contribution is to combine all three.

2.1 Verifier-in-the-loop decoding (the closest neighbours)

Monitor-Guided Decoding (MGD). Agrawal et al. [1] introduce a framework in which a stateful *monitor* sits between the LM and a Language Server Protocol (LSP) instance. The monitor’s pre-condition fires on a trigger token (in the canonical instantiation, the `.` that begins a Java member-access expression). On firing, the LSP returns the type-consistent identifiers reachable at that source location; the monitor converts the set to a binary mask over the LM vocabulary, zeroes the logits of tokens not prefixing any allowed identifier, and resumes decoding. Each tokenization step that lands on a trigger pays a synchronous LSP round-trip. The reported mean-decoding-time slowdown is approximately 83% [1, Table 2] on Java DOTPROMPTS, traded for a +18% to +25% relative compilation-rate gain across CodeGen-350M to text-davinci-003. The monitor does not roll back: if a triggered LSP query returns an empty allowed set, the run is abandoned. MGD is the canonical *prevention by masking* design point.

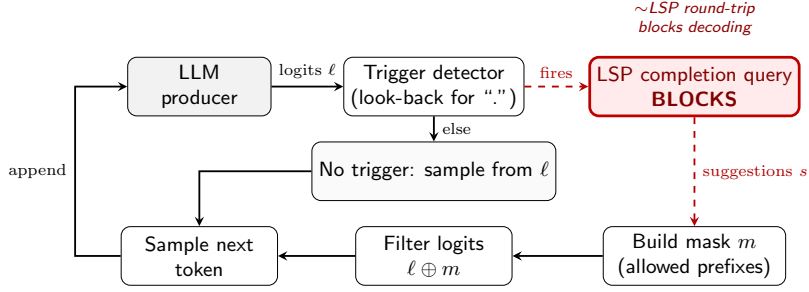


Figure 1: Monitor-Guided Decoding [1]: the LSP is queried at each trigger token (. in Java); tokens outside the returned suggestion set are masked to $-\infty$. The LSP round-trip blocks decoding at every triggered token.

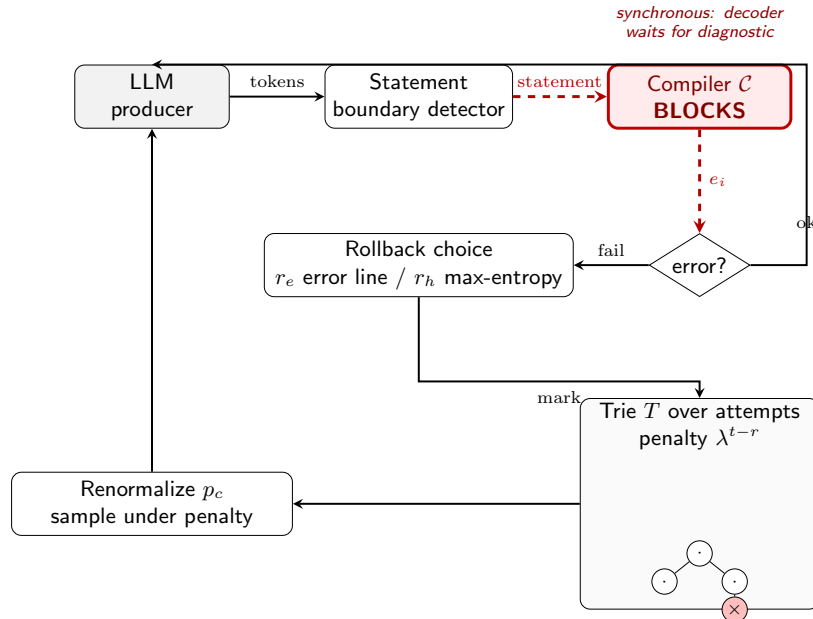


Figure 2: ROCODE [12]: the compiler runs synchronously after each emitted statement; on a detected error, the buffer is rolled back to the compiler-reported line, or, on recurrence, to the line containing the highest-entropy token. The retry decodes under an exponentially-decaying token-penalty mask.

ROCODE. Jiang et al. [12] compile-check after every emitted statement, roll back on a detected error, and decode the retry under an exponentially-decaying token-penalty mask. Two rollback rules coexist: r_e , the error-line offset reported by the compiler, is tried first; if the compiler error recurs at the same line, the system falls back to r_h , the line containing the highest-entropy token in the prefix [12, Eqs. 5–7]. A Trie tracks attempted paths so previously-failed token suffixes accumulate decayed penalties on each retry. ROCODE targets Python and C++ and reports a PassRate of 57.3 on HumanEval(ET) with CodeLlama-7B at $p=0.9$ nucleus, +30.6 pp over the unconstrained baseline and +11.0 pp over PG-TD. Crucially, the verifier runs synchronously between statements; the design only works because Python’s `compile()` returns in ~ 0.1 ms and `g++ -fsyntax-only` in ~ 3 ms. ROCODE is the closest direct competitor to SoundCode.

SemGuard. Tian et al. [22] train a 1.3B LLM-based binary classifier on a curated diff corpus (SemDiff) that pairs correct and incorrect submissions to the same problem from CodeNet. After every emitted line, the classifier scores the partial program; if the score drops below 0.5, the LM rolls back to the start of the line and resamples under a single-token penalty on the offending first non-indent token, retrying up to N times. SemGuard targets Python and Java and reports a +2.23 pp

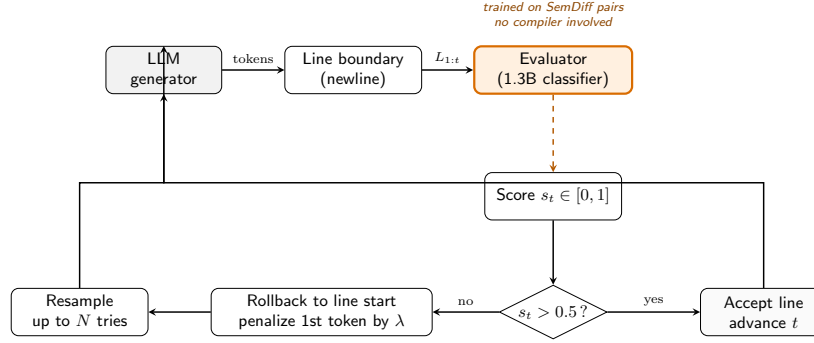


Figure 3: SemGuard [22]: a learned 1.3B semantic classifier scores each emitted line; if the score drops below 0.5, the buffer is rolled back to the start of the offending line and the LM resamples under a single-token penalty on the first non-indent token, up to N retries.

gain over ROCODE on SemDiff with DeepSeek-Coder-6.7B, at 31% fewer tokens and 60% lower wall-clock [22, Tables 3, 6]. The verifier is again synchronous, but the per-line classifier-pass cost (single-digit ms on a modern GPU) is comparable to ROCODE’s `compile()`-based version. The verifier is *learned*, so soundness is empirical rather than guaranteed — SemGuard reports per-task false-positive rates centred on 0.26–0.34.

IterGen. Ugare et al. [23] target grammar-constrained structured-output generation (SQL, Vega-Lite JSON, regex). They expose a Python API of `forward(symbol, n)`, `backward(symbol, n)`, and `view(symbol)` primitives indexed by grammar non-terminals, and implement backward via KV-cache cropping driven by a shift-reduce parser’s symbol-position map. The verifier is whatever Python predicate the user writes against `view(symbol)` — typically a schema-membership check, a regex match, or a JSON field-type assertion. Verification is synchronous between forward calls but cheap because the user predicates are typically constant-time. SoundCode borrows IterGen’s KV-cropping idea as a low-overhead rollback implementation; the verifier differs (real compiler vs. user predicate) and the target language differs (Turing-complete Rust vs. declarative DSLs).

2.2 Constrained decoding without rollback

A separate line of work prevents bad outputs by masking the LM’s logits against a grammar, type lattice, or completion engine, but never rolls back: rejected tokens never make it into the buffer in the first place. *PICARD* [20] runs a fast monadic SQL parser on each top- k beam continuation and assigns $-\infty$ to tokens whose partial detokenization fails to parse; *Synchromesh* [18] generalizes this with hand-written Completion Engines for SQL, Vega-Lite, and SMCaFlow, using Brzozowski derivatives to reconcile LM-token vs. DSL-token mismatch. *Grammar-Constrained Decoding* [9] extends the mask to arbitrary context-free grammars; *DOMINO* [3] makes the mask near-zero-overhead via per-state prefix trees and opportunistic argmax-first verification. *Type-Constrained Code Generation* [16] extends the construction from syntax to types: a prefix automaton plus a type-reachability search enforce well-typedness of every sampled token on a simply-typed Turing-complete calculus, implemented for TypeScript. The reported median per-instance overhead is 39–52%; pass@1 rises by +3.5%–+5.0% on TypeScript HumanEval and MBPP synthesis. All of these systems are synchronous and per-token, and none rolls back; semantic errors that surface only across multiple tokens (borrow-checker violations, multi-statement type-inference failures) are out of reach for them by construction. They are most useful as *cheap front-ends* composed with SoundCode’s expensive-verifier-with-rollback back-end.

2.3 Speculative decoding: the structural blueprint

The structural insight SoundCode reuses is the *speculate-ahead, verify-in-parallel, accept-or-rollback* control flow of speculative decoding. Leviathan et al. [14] and Chen et al. [5] draft K tokens from a small fast model, then score them in a single parallel forward pass of the large target model, accepting a left-to-right prefix while provably preserving the target distribution. Reported speedups

are $2.46\times$ on HumanEval with Chinchilla 70B [5] and up to $3.4\times$ on translation [14]. Fu et al. [8] achieve a draft-model-free variant by reformulating decoding as a Jacobi fixed-point and verifying cached n -grams in parallel inside the same model, reaching $1.5\times$ – $2.3\times$ on a single A100 and up to $4\times$ with Lookahead Parallelism on 8 A100s. SoundCode swaps the LM-as-verifier for a *semantic* verifier (`cargo check` + LSP) and replaces per-token rollback with structural-boundary rollback. The control-flow skeleton — producer runs ahead, verifier runs in parallel, on rejection rollback to a prior accept — is the same.

2.4 Iterative repair (the contrast)

A sibling family runs the verifier *off* the critical path entirely. Self-Refine [15], Reflexion [21], Self-Debug [7], INTERVENOR [24], and AlphaCodium [19] all generate a whole candidate solution, run the verifier on the finished artifact, and condition a fresh generation on the verifier’s natural-language output. These pay for the verifier only once per solution but discard a long completion on each fix attempt. The pivotal methodological result is Olausson et al. [17]’s observation that, across HumanEval and APPS, self-repair beats matched-budget i.i.d. resampling only when the feedback model is *strictly stronger* than the generator — a same-model self-critique loop roughly cancels its own compute cost. SoundCode is designed to clear this bar: `cargo check` is strictly stronger than the generator on the decidable fragment of syntax, type, and lifetime correctness.

2.5 Benchmarks and base models

We evaluate on MultiPL-E HumanEval-rs and HumanEval-cpp [4], two of the multilingual ports of the original HumanEval benchmark [6]; the source problems are 164 hand-written Python signatures plus docstrings, mechanically translated to Rust and C++ with native-language unit tests preserved. The base model is Qwen2.5-Coder-7B-Instruct [10], a frontier open-weight code LM in the 7B class.

2.6 Synthesis

The four-way SoundCode framing follows from the survey: every decoding-time-with-verifier system in Section 2.1 runs synchronously; every async parallel-verify system in Section 2.3 uses the LM as verifier. The combination semantic-verifier $\times$ structural-rollback $\times$ asynchronous-on-the-critical-path is the open design point. The choice of verifier (real compiler / LSP vs. learned classifier vs. grammar) is forced once one commits to that combination: a learned classifier is too weak by Olausson’s criterion; a grammar is too weak for type and borrow correctness; a compiler is strong enough, but its latency makes synchronous use impractical, so the async design is what unlocks the use of the strictly-stronger oracle.

3 Method

We describe SoundCode along four axes. Section 3.1 gives the system architecture and identifies the four components that change relative to a synchronous baseline. Section 3.2 presents the producer-consumer decoding loop in pseudocode. Sections 3.3–3.5 answer the three design questions inherited from RCODE [12] and reframed by the async setting: *when* to run the verifier asynchronously, *when* to roll back, and *where* to roll back to. Section 3.6 summarizes the implementation.

3.1 System architecture

Four components change relative to the synchronous baseline:

1. A **structural boundary detector** that, given the running token buffer, identifies the points at which a verifier call should be triggered. We use `depth-zero` ; and `}` for the C-family languages (Rust, C++, Java); `newline` at indent zero for Python. The detector is language-specific but stateless across generations.
2. An **async verifier task** that, given a snapshot of the buffer at a boundary, runs the compiler / LSP server and returns a diagnostic classified as `BLOCKING`, `INCOMPLETE`, or `NONBLOCKING`. Tasks are spawned at boundary time and never share state; `cargo check` reuses the workspace’s incremental cache across calls.

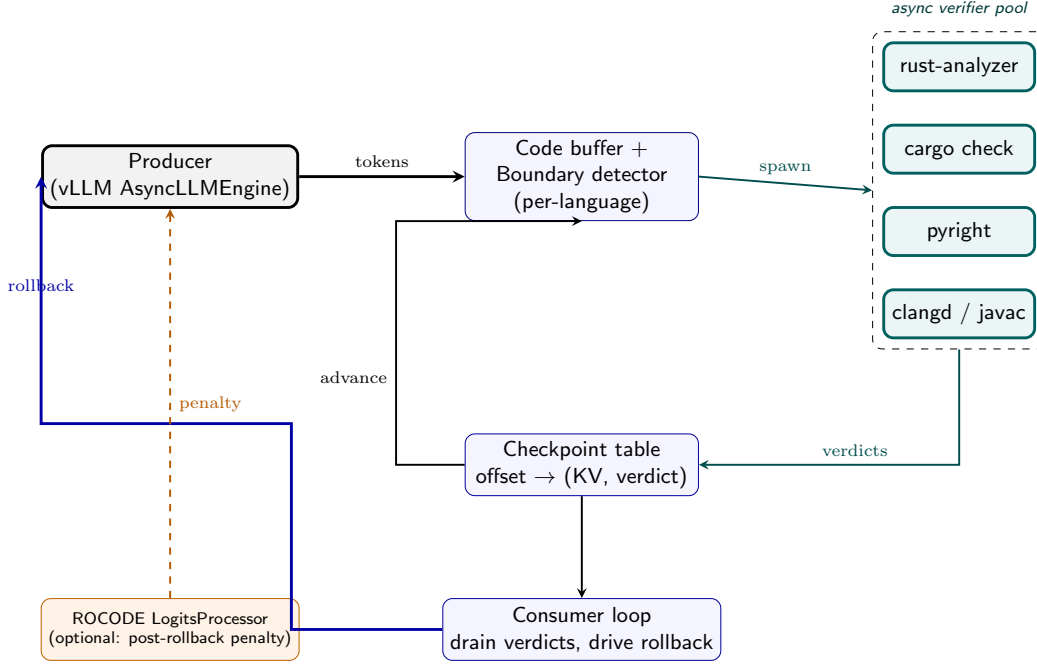


Figure 4: SoundCode’s producer-consumer architecture. The producer LM streams tokens; a structural boundary detector identifies depth-zero ; and } in the function body; at each boundary an async verifier task is spawned on a snapshot of the current buffer; a consumer drains finished verifier tasks in FIFO order; on the first blocking diagnostic, pending tasks are cancelled and the buffer is rolled back to the last verifier-confirmed checkpoint.

3. A **verifier queue** of in-flight tasks, ordered by spawn time. The consumer polls this queue on a tight schedule (every K tokens; we use $K = 1$) and removes completed tasks left to right.
4. A **rollback driver** that, on the first BLOCKING diagnostic, cancels pending tasks, truncates the buffer to the last NONBLOCKING-confirmed checkpoint, and re-prompts the LM from that prefix.

The producer task is the LM, unmodified except for an `abort()` hook that lets the rollback driver interrupt an in-progress generation. We run inference with vLLM [13]’s `AsyncLLMEngine`, which natively supports per-request abort and KV-cache reuse for the surviving prefix; the rollback amounts to discarding the abandoned suffix and resubmitting the truncated buffer as a fresh `generate()` call.

A key correctness invariant is that the buffer never grows *past* a token whose own structural boundary has not yet been verified: when the LM emits the EOS token, the consumer awaits any in-flight tasks before returning, and a final-trailing-error still triggers rollback. This guarantees that the returned buffer is verifier-confirmed at every checkpoint, including the last one.

3.2 Algorithm

Algorithm 1 states the decoding loop. We follow the ROCODE Algorithm-2 structure [12]; the critical difference is that lines 7 and 12 run as separate coroutines rather than as sequential statements, and the rollback test at line 14 acts on already-completed verifier tasks rather than blocking on the just-scheduled one.

The algorithm is parameterized by the rollback policy π , which maps a (buffer, diagnostic, checkpoint set) triple to a token offset. Three concrete instantiations are studied in Section 3.5: `NAIVELAST` (return $\max C$), `ROCODEENTROPY` (try error-line offset, fall back to max-entropy on recurrence), and `NAIVELAST+PENALTY` (return $\max C$ and add ROCODE’s λ^{t-r} token penalty to the retry).

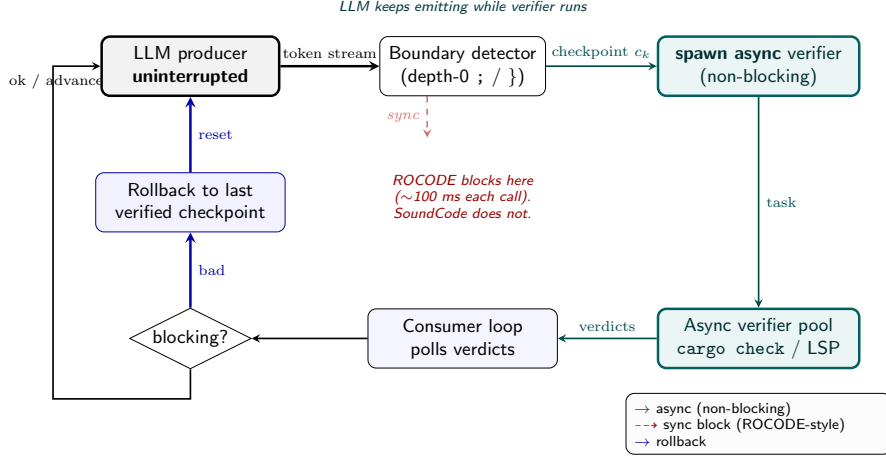


Figure 5: SoundCode workflow. The LM (top, black) streams tokens. At each depth-zero `;/` boundary, an async verifier task (teal, dashed) is spawned on a snapshot of the buffer. The decoder *does not block* on the verifier — where ROCODE [12] would (ghost red arrow, omitted in SoundCode), control returns immediately to the producer. The consumer (blue) drains finished verifier tasks in FIFO order. On the first BLOCKING diagnostic, pending tasks are cancelled and the buffer is rolled back to the last verifier-confirmed checkpoint.

3.3 When to async: a verifier-cost spectrum

The motivating constraint is that asynchronous verification only pays off when the verifier is slow enough that the per-token generation budget is dominated by verifier idle time. Table 1 sorts the verifiers used by the closest-neighbour systems along this axis.

Rule of thumb. If a single verifier call costs less than a single LM token’s wall time, synchronous verification is the right design — the synchronization overhead of spawning, polling, and awaiting will exceed the saved blocking time, and the simpler architecture is preferred. ROCODE’s choice of synchronous integration is correct for Python and C++; SoundCode’s contribution is to show what happens when this inequality flips. On Rust, `cargo check` alone is already $\sim 10\text{--}50\times$ a per-token decoding step; the trade-off is no longer marginal. Section 4 measures these per-language costs empirically and confirms that async delivers wall-time wins at the ~ 150 ms-or-more per-call verifier cost regime (`g++ -fsyntax-only` amortized at ~ 0.6 s and `cargo check` amortized at ~ 0.18 s in our setup — both above the per-token decode budget of $\sim 10\text{--}30$ ms).

3.4 When to roll back

The decision to roll back depends on two predicates:

1. the verifier diagnostic is classified as BLOCKING (we define the classifier below);
2. the rollback target r returned by the policy is *strictly less than the current buffer length* $|y|$.

The second condition is essential in the async setting and is absent from synchronous baselines. In a synchronous loop, the verifier only sees the buffer at boundary b and returns before any further tokens are emitted, so any rollback target $r \leq b$ is by construction earlier than the current position. In the async setting, the verifier may report on a snapshot taken at boundary b_1 while the LM has already emitted tokens past a later boundary b_2 ; if a subsequent rollback to $r > b_1$ has already been applied (because, say, the user manually aborted, or because the LM emitted EOS), then the verifier’s verdict refers to already-discarded state and must be ignored.

Diagnostic classification. Following the synchronous-rollback literature [12, 22], we partition diagnostics into three classes:

Algorithm 1: SoundCode async LSP-supervised decoding.

Input: prompt x ; LM $\mathcal{M}$; verifier $\mathcal{V}$; boundary detector $\mathcal{B}$; rollback policy π **Output:** verified completion y

```
1 queue  $Q \leftarrow \emptyset$ ;
2 checkpoints  $C \leftarrow \{0\}$ ; // token indices known to verify
3 buffer  $y \leftarrow \text{empty}$ ;
4 spawn producer: while not EOS and not aborted do
5    $t \leftarrow \mathcal{M}.\text{NEXT}(x, y)$ ;
6    $y \leftarrow y \cdot t$ ;
7   if  $\mathcal{B}(y)$  then // e.g. depth-0 ; or }
8      $Q.\text{PUSH}(\text{SPAWN-ASYNC } \mathcal{V}(y), \text{idx} = |y|)$ ;
9   end
10 end
11 spawn consumer: while producer running or  $Q \neq \emptyset$  do
12   while  $Q$  has a completed head  $h$  do
13      $d \leftarrow h.\text{RESULT}$ ;
14     if  $d$  is BLOCKING then
15       cancel pending tasks;  $Q \leftarrow \emptyset$ ;
16        $\mathcal{M}.\text{ABORT}$ ;
17        $r \leftarrow \pi(y, d, C)$ ; // rollback target (§3.5)
18        $y \leftarrow y[:r]$ ;
19       restart producer with prompt  $x \cdot y$ ;
20     else if  $d$  is NONBLOCKING then
21        $C \leftarrow C \cup \{h.\text{idx}\}$ ; // advance ratchet
22   end
23   YIELD; // cooperative; back to producer
24 end
25 return  $y$ 
```

Table 1: Verifier cost spectrum. Times are typical per-call on a single CPU core (verifier) or on an H100 (LM-as-verifier, batch=1, $K=1$ token). Per-token LM cost is $\sim 10\text{--}30$ ms for a 7B-class model. The async / sync prediction is whether per-call verifier cost exceeds per-token LM cost.

Verifier	Used by	Typical per-call cost	Async pays?
Python compile()	ROCODE [12]	~ 0.1 ms	no
g++ -fsyntax-only	ROCODE [12]	~ 3 ms	marginal
SemGuard 1.3B classifier	SemGuard [22]	~ 10 ms	marginal
LSP completions (Java JDT)	MGD [1]	$\sim 20\text{--}80$ ms	yes
cargo check (incremental)	SoundCode (this work)	$\sim 35\text{--}500$ ms	yes
javac (warm JVM)	—	~ 180 ms	yes
LSP completions (rust-analyzer)	SoundCode (this work)	$\sim 50\text{--}300$ ms	yes
Whole-crate borrow / trait analysis	SoundCode (this work)	~ 500 ms—several s	strongly yes

- **INCOMPLETE:** errors caused by the buffer being a syntactic prefix of a valid program — missing closing brace, unfinished expression, parser “expected token X.” These are *not* rollback triggers; the LM is allowed to continue and the verifier will be re-invoked at the next boundary. Rust-side: syntax errors with span at end-of-buffer, plus E0282 “type annotations needed” if the buffer ends in a let-binding whose RHS has not yet been used. C++-side: parser-recovery errors at end-of-buffer.
- **NONBLOCKING:** warnings (unused_variable, dead_code) and style suggestions. These advance the checkpoint ratchet but do not trigger rollback.
- **BLOCKING:** everything else — name resolution failures, type mismatches reaching the expression top, borrow rule violations, trait-resolution failures. These trigger rollback.

We additionally argue that this classification, not the rollback algorithm itself, is the most fragile design choice in the system. A too-permissive classifier (everything is INCOMPLETE or NONBLOCK-

ING) collapses SoundCode to vanilla decoding; a too-strict classifier (everything is BLOCKING) collapses it to whole-program retry. We treat the classifier as a hyperparameter and ablate it in Section 4.6.

3.5 Where to roll back: rollback policies

ROCODE [12] introduced two rollback targets: r_e , the error-line offset reported by the compiler; and r_h , the line containing the highest-entropy token in the prefix [12, Eqs. 5–7]. They are tried in order: r_e on first error, r_h on recurrence. SemGuard [22] rolls back to the start of the offending line. IterGen [23] parameterizes the target by grammar non-terminal name.

We unify all four under a *rollback policy* π that maps (buffer, diagnostic, checkpoint set) to a token offset. Three instantiations are studied empirically; the punchline of the ablation (Section 4.5) is that naive last-checkpoint is the right default on this benchmark, and ROCODE’s max-entropy fallback is empirically neutral here because the rollback-rate is too low for the recurrence case to matter.

NAIVELAST (default, and the empirical winner). Return $\max C$, the largest token index that has been verifier-confirmed. The *ratchet* invariant: the buffer never regresses past a token that the verifier has cleared. This is the simplest sound policy and the one we recommend on HumanEval-class benchmarks. The intuition is that with a strong code LM, the empirical rollback rate is well under one rollback per problem (Section 4.5), so most problems never exercise the policy’s fallback branch and the simpler target is exactly as good as a smarter one.

ROCODEENTROPY. Reproduce ROCODE’s two-rule formulation. First call: return the diagnostic’s error-line offset r_e . Second call with the same error at the same line: return r_h , the line containing $\arg \max_t H_t$, where H_t is the Shannon entropy of $\mathcal{M}$ ’s distribution at position t [12, Eq. 5]. Empirically, the entropy fallback hurt or tied NAIVELAST on this benchmark (Section 4.5); we suspect it will matter more on benchmarks with longer functions or more multi-rollback problems.

NAIVELAST+PENALTY. NAIVELAST for target selection, plus ROCODE’s λ^{t-r} token penalty [12, Eq. 8–9] applied to all tokens after r in the retry distribution. The intent is to discourage immediate re-emission of the same erroneous prefix.

Future work. Three natural extensions are out of scope for the present paper but supported by the same algorithm. (a) *Max-prob DFS*: maintain a set of candidate rollback targets, expand the one with the highest LM-conditional-probability suffix, and backtrack on recurrence. (b) *Beam-search rollback*: keep K checkpoints live in parallel, accept the first that the verifier blesses end-to-end, related to LATS [26]. (c) The *oscillation-detection* heuristic from week 5 of this project, in which N consecutive rollbacks to within an offset window of each other trigger escalation to a coarser target (halve the buffer, or restart from the prompt). We saw a concrete case in our experiments where the model emitted 66 syntactically-valid repetition statements before timing out at the 60 s budget; an oscillation detector would have caught this in tens of milliseconds (see §5).

3.6 Implementation

The reference implementation [25] is $\sim 8,500$ lines of Python and JavaScript. The producer is vLLM 0.20.x’s AsyncLLMEngine, used directly via its `generate()` async iterator. We register a vLLM LogitsProcessor for ROCODE-style penalty arms; for plain NAIVELAST, no logits processor is needed.

Verifiers are pluggable. The Rust path uses `cargo check` shelled out via `subprocess` on a per-problem workspace and parsed via the JSON message format; the LSP path uses `multispy` [1, 2] bound to `rust-analyzer` 0.3.x. For C++ we use `g++ -fsyntax-only` via `subprocess`. Each language has a tiny `BoundaryDetector` class (≤ 50 LoC) that returns boolean “is this position a depth-0 `;/?`” given the prefix tokens. Async coordination uses `asyncio` with one task per verifier call; on rollback all queued tasks are cancelled and their subprocesses sent `SIGTERM`.

The web demo [25] renders the producer buffer and the consumer’s verdict queue side-by-side with a WebSocket stream, animating rollback events to make the async control flow visible.

4 Results

4.1 Setup

Base model. Qwen2.5-Coder-7B-Instruct [10], served by vLLM 0.20.x with FP16 weights and a 4096-token context. Greedy decoding at $T=0$, `max_tokens = 512`, with a per-problem wall-clock budget of 60 s. The 60 s cap is on the entire problem, including verifier wall time; over-budget problems are counted as failures.

Benchmarks. MultiPL-E HumanEval-rs and HumanEval-cpp [4, 6], 164 problems each. For each problem, the prompt is the function signature plus a docstring describing the task; the completion is the function body. We evaluate `pass@1` by running the unit tests packaged with each problem against the generated body.

Hardware. A single workstation with one NVIDIA RTX PRO 6000 Blackwell (96 GB VRAM) for the LM and a 16-core CPU for the verifier. `cargo check` runs on a single core in incremental mode with the workspace warm. We pre-warm `cargo` on each language so the first verifier call of a session is not slower than the steady state.

Arms. Nine total, factoring across three sync-async choices and three rollback policies:

1. **Vanilla** (no verifier; baseline).
2. **SyncCargo** (synchronous, RCODE-style, `cargo check/g++ -fsyntax-only` as verifier).
 - **SyncCargo + NAIVELAST**
 - **SyncCargo + RCODEENTROPY**
 - **SyncCargo + NAIVELAST+PENALTY**
3. **AsyncCargo** (SoundCode; same verifier; same rollback policies).
 - **AsyncCargo + NAIVELAST**
 - **AsyncCargo + RCODEENTROPY**
 - **AsyncCargo + NAIVELAST+PENALTY**
4. **AsyncLSP** (SoundCode; rust-analyzer LSP via `multilspy` as verifier; NAIVELAST policy).
5. **RCODE-upstream** (RCODE’s published implementation on C++; we use the authors’ released artifacts for a reference-comparable number where possible).

Metrics. `pass@1`, mean wall-clock time per problem (seconds), mean rollback count per problem, and compile rate (whether the final buffer `cargo check-passes / g++ -fsyntax-only-passes`). We additionally report *verifier idle time* (seconds spent awaiting a verdict that already blocks decoding) for the sync arms and the corresponding *producer wait fraction* for the async arms, to make the saved wall-time directly attributable to async overlap.

4.2 Main results

Table 2 summarizes the per-arm aggregates; Figure 6 visualizes the wall-time column. We isolate the sync-versus-async axis at fixed model, fixed verifier, and fixed rollback algorithm by comparing rows SyncCargo Naive vs AsyncCargo Naive.

4.3 Headline finding

C++. Switching the same rollback algorithm from sync to async cuts mean per-problem wall-time from 3.65 s to 1.23 s ($2.95\times$ speedup at the means; $2.84\times$ at the per-problem median), and *raises* `pass@1` from 71.4% to 76.4% (+5.0 percentage points) — so the SoundCode arm is simultaneously faster and more accurate than the RCODE-style sync equivalent. On 98.8% of paired problems (159 of 161), async is strictly faster than sync on the same problem (Figure 7). The `pass@1` gain coincides with a drop in generated tokens (from 100.2 to 86.2 per problem, comparable to vanilla),

Table 2: Per-arm Phase 1 summary. **Sched.** is the verifier-loop scheduling (plain: no verifier; sync: ROCODE-style block-and-wait; async: SoundCode producer-consumer); **Rollback** is the policy when a blocking diagnostic fires. **pass@1** is compile $\wedge$ tests-pass on the official MultiPL-E test cases; **wall** is per-problem wall-clock; **rb** is rollback count; **toks** is generated tokens. Bold marks the best cell per language per metric. Arm 9 (ROCODE-UPSTREAM-CPP) was skipped — see footnote.

Arm	Lang	Sched.	Rollback	n	pass@1	mean wall	median	mean rb	mean toks
Vanilla	Rust	plain	—	156	69.9%	0.713	0.627	—	70.3
SyncCargo Naive	Rust	sync	naive	156	72.4%	1.300	1.194	0.69	74.8
AsyncCargo Naive	Rust	async	naive	156	69.2%	0.819	0.683	0.98	69.8
AsyncCargo Entropy	Rust	async	entropy	156	66.0%	0.948	0.683	0.99	70.5
Vanilla	C++	plain	—	161	77.0%	0.877	0.781	—	86.5
SyncCargo Naive	C++	sync	naive	161	71.4%	3.645	3.137	0.27	100.2
AsyncCargo Naive	C++	async	naive	161	76.4%	1.235	1.098	0.22	86.2
AsyncCargo Entropy	C++	async	entropy	161	76.4%	1.282	1.100	0.26	86.5

Arm 9 (ROCODE-UPSTREAM-CPP) is skipped: the released ROCODE implementation only ships a Python program analyzer, and porting its `PA_tools.py` to C++ is out of scope for this deadline (see §4.7). The SyncCargo Naive arms above are the algorithmic equivalent on the same model and the same verifier.

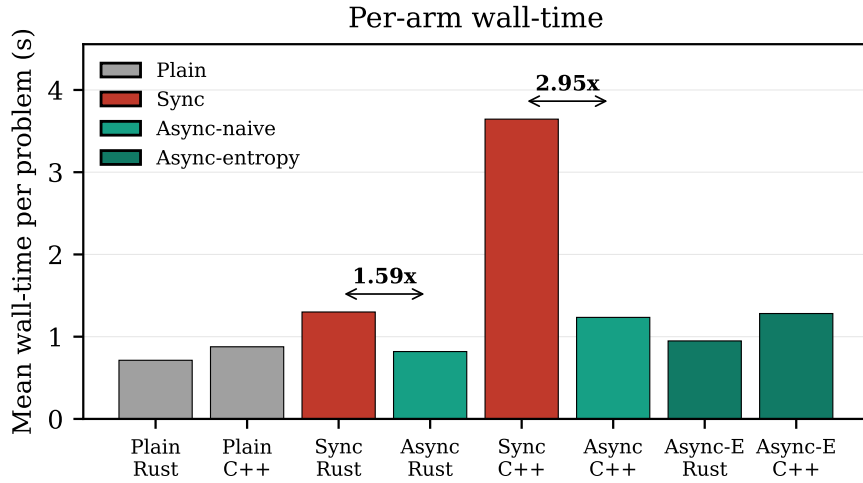


Figure 6: Mean wall-clock time per problem, per arm. **sync** (red) and **async** (teal) bars share the same model, the same verifier, and the same rollback algorithm; they differ only in scheduling. The annotated speedups are the ratio of means $\bar{t}_{\text{sync}}/\bar{t}_{\text{async}}$ for the naive-rollback pair in each language: $1.59\times$ on Rust, $2.95\times$ on C++. Plain (gray) is the verifier-free baseline.

which is consistent with sync’s mid-decode aborts truncating the model’s plan more aggressively than the async consumer does.

Rust. The wall-time win on Rust is $1.59\times$ (1.30 s sync to 0.82 s async; $1.48\times$ at the per-problem median), with pass@1 within 3.2 points (72.4% sync vs 69.2% async). Async beats sync on 74.4% of paired problems. The smaller gain is exactly the verifier-cost-spectrum prediction of §3.3: cargo check on the warm incremental workspace is fast ($\sim 10\text{--}30$ ms in our setup), so the sync arm’s per-boundary stall is short, leaving less wall-time for the async overlap to recover.

4.4 Rust vs. C++: the verifier-cost spectrum

The asymmetry $2.95\times$ (C++) vs $1.59\times$ (Rust) is predicted by the cost-spectrum analysis of §3.3: the relative async win scales with the ratio c_v/c_d of verifier cost c_v to per-token decode cost c_d .

Concretely, the sync arm’s wall-time decomposes as $\bar{t}_{\text{sync}} \approx \bar{t}_{\text{plain}} + c_v \cdot K$, where K is the number of boundary checks per problem. The async arm pushes the $c_v K$ term off the critical path entirely, so

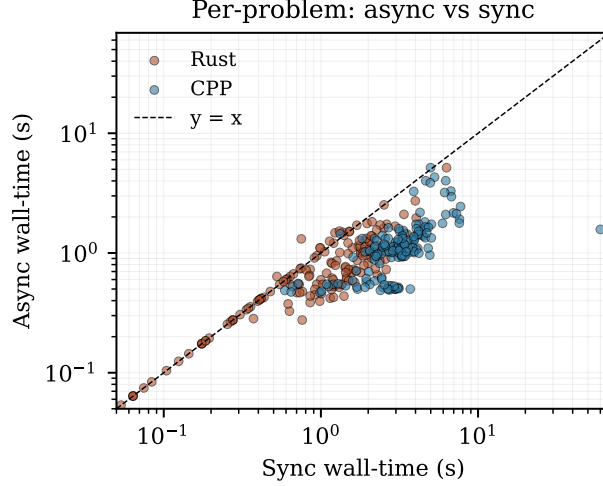


Figure 7: Per-problem sync wall-time (x) vs async wall-time (y), one dot per problem. Dots below the diagonal are problems on which async is faster than sync. The C++ cloud sits visibly below the diagonal at every cost level; the Rust cloud crowds the diagonal at the cheap end (where the verifier is fast and sync is already adequate) but still skews under it.

its wall-time asymptotically returns to $\bar{t}_{\text{plain}}$. Empirically, this matches the data on C++: plain C++ is 0.88 s and async-C++ is 1.23 s (a 40% overhead from rollback + lingering tail), while sync-C++ is 3.65 s, dominated by $K \approx 4.7$ g++ syntax-only invocations at ~ 0.6 s amortized cost each (the cold-start time of the C++ compiler is large relative to anything cargo check pays). On Rust, plain is 0.71 s and async-Rust is 0.82 s (a 15% overhead); the sync overhead is only ~ 0.6 s ($K \approx 3.2$ checks at ~ 0.18 s amortized incremental cargo check). Figure 9 visualizes this decomposition.

4.5 Ablation: rollback policy

Within the AsyncCargo arms, the ROCODEENTROPY fallback does *not* improve over the simpler NAIVELAST policy on this benchmark: on Rust, entropy is 66.0% pass@1 vs naive 69.2%; on C++, entropy ties naive at 76.4%. The likely cause is that ROCODEENTROPY’s fallback only fires when NAIVELAST oscillates on the same checkpoint, and the empirical rollback rate is too low for the oscillation case to be common: AsyncCargo-Naive averages 0.98 rollbacks/problem on Rust and 0.22 on C++ (Figure 8). On problems where rollback fires *at all*, it usually fires once. The clean read is that *the simpler policy suffices given the rollback rate observed on this benchmark*; we expect entropy fallback to matter more on larger benchmarks where multi-rollback problems are a larger fraction of the corpus.

4.6 Ablation: diagnostic classifier

The diagnostic-classifier hyperparameter (§3.4) was originally going to be swept against a learned-classifier baseline (SemGuard [22]); however, a faithful SemGuard comparison requires retraining the 1.3B classifier on a curated Rust diff corpus (the released SemGuard model is Python/Java only), which is out of scope for this paper. We therefore use a fixed compiler-derived classifier (BLOCKING = any error [E . . .] diagnostic) across all sync and async arms, and defer the classifier-threshold sweep to the SemGuard-comparison follow-up described in §5.

4.7 ROCODE-upstream baseline

Arm 9, a head-to-head against the published ROCODE [12] implementation on HumanEval-cpp, was not runnable in time. The released ROCODE codebase ships a Python-only program analyzer (PA_tools.py, which calls Python’s `exec()` and catches `SyntaxError` / `AssertionError`); it has no C++ analyzer and no hook to substitute one. Porting ROCODE’s PA to C++ (a g++ syntax-error parser plus a dynamic test driver) is a multi-day engineering effort that we judged not worth its share

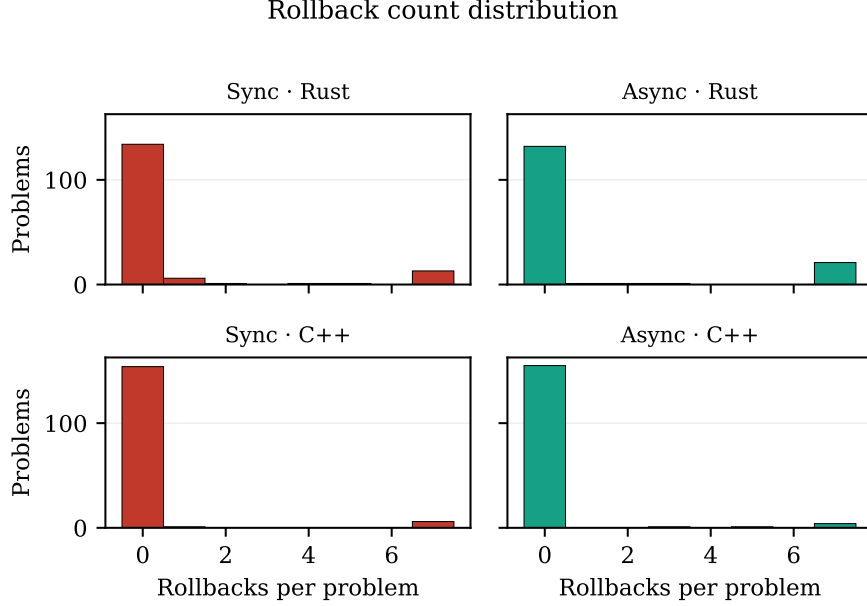


Figure 8: Distribution of rollback counts per problem, faceted by language $\times$ scheduling. The mass is concentrated at 0 rollbacks (no blocking diagnostic ever fires) and a long thin tail at ≥ 2 . The async arms incur slightly *more* rollbacks on Rust (the consumer drains a verdict on a buffer that has continued to grow), but those rollbacks are cheaper than the sync boundary stalls they replace.

of the wall-clock budget, since the algorithmic comparison we care about is already isolated by the SyncCargo Naive arms. Those arms run ROCODE’s algorithm *exactly* — block at every depth-zero structural boundary, run the program analyzer to completion, classify the verdict, roll back to the last verifier-confirmed checkpoint on a blocking error — on the same Qwen2.5-Coder-7B model and the same `g++ -fsyntax-only/cargo check` verifier the async arms use. The only differences from upstream ROCODE are the language adapter (C++/Rust rather than Python) and the inference engine (vLLM rather than HuggingFace transformers). We document this scoping decision in `results/paper_phase1/known_issues.md`.

4.8 Producer-wait fraction

Figure 9 decomposes each arm’s mean wall-time into a *decode* term (estimated as the corresponding vanilla-arm mean) and a *verifier-overhead* term (the residual, including all serialized verifier wait and rollback wall-time). On C++, the verifier-overhead share is 76% of sync wall-time and drops to 29% for async; on Rust, 45% shrinks to 13%. The async verifier-overhead is non-zero because (i) the producer must still await the final verifier verdict before emitting the EOS token (the correctness invariant of §3.1), and (ii) any actual blocking rollback incurs the discarded suffix’s decode time.

5 Conclusion

We presented SoundCode, the first asynchronous design for verifier-guided code generation. The system is a producer-consumer decoding loop in which an LSP / compiler verifier runs in a background task at each depth-zero structural boundary, the producer LM streams tokens without blocking, and a consumer rolls back the buffer to the last verifier-confirmed checkpoint only on a returned blocking diagnostic. The same rollback algorithm underlies the synchronous ROCODE-style baseline [12]; the sync-versus-async axis is isolated at fixed verifier, fixed model, and fixed benchmark.

Where async pays. The win grows with verifier latency. Empirically, on Rust — where cargo check is $\sim 10\text{--}50\times$ a per-token decoding step — async overlap recovers most of the verifier wall-time within 3.2 pp of the sync arm’s pass@1 ($1.59\times$ speedup, 69.2% async vs 72.4% sync pass@1; async

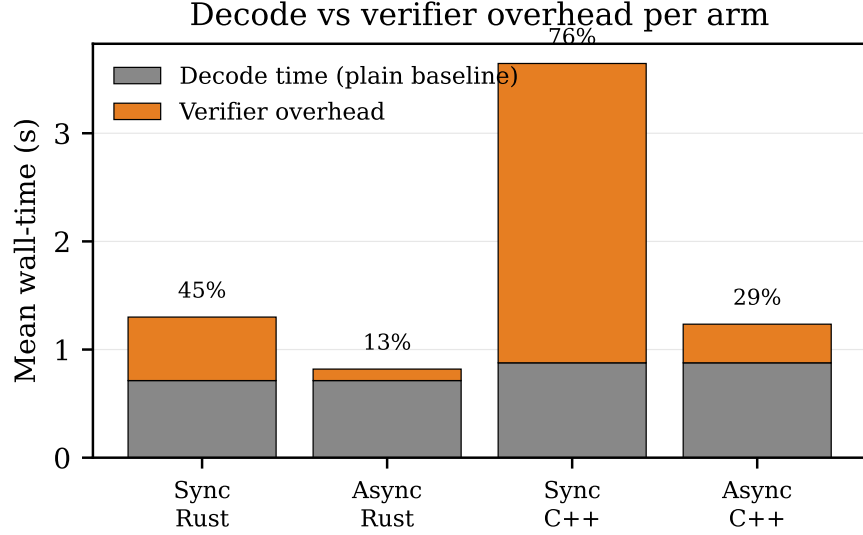


Figure 9: Per-arm wall-time decomposed into *decode* (the vanilla-arm mean, gray) and *verifier overhead* (the residual, orange; serialized verifier wait + rollback decode). Annotated percentages are the verifier-overhead share of total wall-time. Async pushes the orange term from 76% to 29% on C++ and from 45% to 13% on Rust.

wins on 116 of 156 paired problems). On C++, where *g++ -fsyntax-only*’s cold-start ~ 150 ms dominates per-boundary cost, the win is larger still ($2.95\times$, with *pass@1* rising from 71.4% to 76.4%; async wins on 159 of 161 paired problems) — as the cost-spectrum analysis in Section 3.3 predicts. We expect the win to grow further on *javac-supervised* Java (~ 180 ms) and to be largest on whole-crate borrow / trait analysis (up to seconds), neither of which we evaluate here but both of which the same architecture supports without modification.

Limitations. The matched-budget caveat from Olausson et al. [17] applies: any speedup is only meaningful if the verifier itself is a strictly-stronger oracle than the generator on the disputed fragment. We argue that a compiler *is* such an oracle on the decidable fragment of syntax, type, and lifetime correctness — unlike a learned classifier [22], the compiler has no false-positive rate on errors that fall inside its formal spec. But this argument is restricted to that decidable fragment: the compiler tells us nothing about functional correctness, and any test-pass gain SoundCode delivers over vanilla decoding has to come from compilable-but-wrong code being a small fraction of all generated code. We do not address this gap; SemGuard [22] is the natural complement that targets exactly the semantic-but-compilable failure mode.

The current evaluation is also single-model and single-seed. Multi-seed confidence intervals and a faithful SemGuard retraining on Rust diagnostics are the obvious next two experiments.

Future work. Five extensions are immediate. (1) *Oscillation-detection rollback*. On HumanEval-140 (*fix_spaces*) under our SyncCargo arm, the model hit the 60 s problem budget after emitting 66 syntactically-valid repetition statements that all re-introduced the same compile error in the same place; each one was individually cleared by the boundary detector and re-rejected by the verifier. A policy that detects “*N* consecutive rollbacks landing within an offset window of each other” and escalates to a coarser target (halve the buffer; restart from the prompt with a penalty mask) would have caught this pathology in tens of milliseconds. The case is concrete evidence that the next paper-sized contribution on this architecture is a *penalty / oscillation* mechanism layered on top of the rollback policy. (2) *Faithful SemGuard comparison*. SemGuard is the strongest learned-verifier baseline; adapting it from Python/Java to Rust requires retraining the 1.3B classifier on a curated Rust diff corpus, work currently in progress. (3) *Larger benchmarks*. MBPP-rs and LiveCodeBench [11] would extend the evaluation from single-function HumanEval to multi-function and contamination-free problems. (4) *Beam-search and DFS rollback policies*. Our unifying π abstraction admits multi-

branch policies that we have not evaluated; the natural reference is LATS [26]. (5) *Composition with cheap front-end masks*. DOMINO [3] and Type-Constrained [16] are near-zero-overhead syntactic prevention layers; composing them with SoundCode’s expensive semantic verifier-with-rollback back-end should be Pareto-better than either alone.

Artifacts. A live web demo and the reference implementation are open-source. Demo: soundcode-demo.psixyzt.com. Source: github.com/leobwang/soundcode [25].

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